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Modulation of Human iPSC-derived Hepatocyte Phenotype Via Extracellular Matrix Microarrays

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Abstract

In vitro human liver models are essential for drug screening, disease modeling, and cell-based therapies. Induced pluripotent stem cell (iPSC)-derived hepatocyte-like cells (iHeps) mitigate sourcing limitations of primary human hepatocytes (PHHs) and enable precision medicine; however, current protocols yield iHeps with very low differentiated functions. The composition and stiffness of liver's extracellular matrix (ECM) cooperatively regulate hepatic phenotype *in vivo*, but such effects on iHeps remain unelucidated. Here, we utilized ECM microarrays and high content imaging to assess human iHep attachment and functions on ten major liver ECM proteins in single and two-way combinations robotically spotted onto polyacrylamide gels of liver-like stiffnesses; microarray findings were validated using hydrogel-conjugated multiwell plates. Collagen-IV supported higher iHep attachment than collagen-I over 2 weeks on 1 kPa, while laminin and its combinations with collagen-III, fibronectin, tenascin C, or hyaluronic acid led to both high iHep attachment and differentiated functions; laminin and its combination with tenascin or fibronectin led to similar albumin expression in iHeps and PHHs. Additionally, several collagen-IV-, laminin-, fibronectin-, and collagen-V-containing combinations on 1 kPa led to similar or higher CYP3A4 staining in iHeps than PHHs. Lastly, collagen-I or -III mixed with laminin, collagen-IV mixed with lumican, and collagen-V mixed with fibronectin led to high and stable functional output (albumin/urea secretions; CYP1A2/2C9/3A4 activities) in iHep cultures versus declining PHH numbers/functions for 3 weeks within multiwell plates containing 1 kPa hydrogels. Ultimately, these platforms can help elucidate ECM's role in liver diseases and serve as building blocks of engineered tissues for applications.

Keywords

iPSC technology; extracellular matrix; drug screening; cytochrome P450

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1. INTRODUCTION

Due to significant species-specific differences in drug metabolism pathways, *in vitro* human liver models are critical for assessing the metabolism and toxicity of chemicals before human exposure [1] and for discovering drugs targeted against liver diseases (e.g., alcoholic and non-alcoholic fatty liver diseases and hepatitis B/C viral infections) [2]. While primary human hepatocytes (PHHs) are the ‘gold standard’ for constructing human liver models due to their physiological relevance and the ability to function for several weeks *in vitro* under the appropriate culture conditions [3], the scarcity of healthy liver tissue limits PHH use for high-throughput studies, including drug screening, and a lack of available PHH donor diversity makes it difficult to explore the genetic basis of diseases. In contrast, induced pluripotent stem cells (iPSCs) can address PHH limitations and serve as a sustainable and genetically-diverse cell source for applications [4]. Several *in vitro* protocols relying on soluble factors to mimic the various stages of liver development *in vivo* have been established to produce iPSC-derived human hepatocyte-like cells (iHeps); however, these protocols are not able to fully differentiate iHeps into an adult phenotype, especially with respect to drug metabolism capacity [4, 5], which limits the use of these cells for drug screening, disease modeling, and regenerative medicine.

The human liver has an extracellular matrix (ECM) of complex composition and stiffness in the ≤ 4 kPa Young’s modulus range, as measured by atomic force microscopy or ultrasound imaging [6, 7]. *In vitro*, the iHep phenotype has been shown to be sensitive to the ECM protein composition and the substrate stiffness, albeit separately. For instance, Matrigel™ is used to differentiate iHeps [8, 9], but this ECM is limited by batch-to-batch variability in its protein concentration and composition. As an alternative, human liver ECM (hLECM) has been utilized to differentiate iHeps *in vitro* [10], but like Matrigel, hLECM suffers from batch-to-batch irreproducibility in protein content due to the variable conditions of sourced liver tissues. Furthermore, it is not clear which components within complex ECM gels synergistically regulate different iHep fates. In contrast, recombinant ECM proteins are useful to mitigate the above limitations with complex ECM gels. For instance, LN 411 and 111 isoforms upregulate albumin and cytochrome P450 (CYP) enzyme expression, respectively [11, 12]. With respect to substrate stiffness, mouse embryonic stem cell (ESC)-derived [13] and human ESC-derived hepatocyte-like cells [14] produced higher levels of albumin on hydrogel substrates with liver-like stiffness versus on stiffer substrates. However, it remains unclear how ECM stiffness synergizes in unexpected ways with its protein composition to regulate iHep maturation, knowledge that can be useful to engineer next generation platforms for drug development and regenerative medicine.

Screening for the combinatorial effects of ECM stiffness and protein compositions on cellular phenotype using multi-well plates can become cost-prohibitive, especially across thousands of combinations and different tissue donors. In contrast, high-throughput ECM microarrays allow for the independent modulation of cell-cell interactions, ECM protein composition, substrate stiffness, and overlaid soluble factors [15–17]. In this microarray platform, a robotic spotter is used to deposit ECM proteins onto a dehydrated polyacrylamide (PA) hydrogel conjugated onto a glass slide. The proteins are then retained in the hydrated PA gel to facilitate cell attachment while the surrounding hydrophilic

PA resists protein adsorption and thus cell adhesion. An automated high-content imaging pipeline is used to obtain single-cell level data from microarrays to generate quantitative comparisons of the effects of combinatorial microenvironments on cell phenotype [16].

We previously used ECM microarrays to show that ECM protein composition and soluble factors synergize in novel ways to affect the differentiation of mouse progenitor cells [18, 19], the functions of PHHs [20], and the phenotype of hepatic stellate cells in the context of liver fibrosis [21]. Here, we sought to use ECM microarrays to elucidate for the first time the combinatorial effects of ten major ECM proteins present in the liver and liver-like substrate stiffnesses on the attachment and phenotypic protein staining patterns of adult-like (albumin and CYP3A4) and fetal-like (alpha-fetoprotein or AFP) markers in human iHeps from multiple donors over 2 weeks in culture, while using collagen-I as the control ECM utilized commonly for iHep culture [22]; iHep responses on microarrays were compared against published PHH data on similar ECM microarrays [20]. Lastly, we validated select ECM regulators of iHep staining patterns obtained from the microarrays within multiwell plates containing hydrogels of different stiffnesses, which also allowed us to evaluate secreted markers (albumin and urea) and the metabolism of different CYP enzyme substrates, with comparisons to PHH responses in similar plates.

2. METHODS

2.1 Hydrogel preparation and microarray fabrication

PA gels were selected due to their biocompatibility with human cells, tunability of gel stiffness via crosslinking ratios, amenability to selective ECM deposition, and previous validation for use with cellular microarrays [18, 23]. Notably, PA-based cellular microarrays have been fabricated and validated for human liver cells as previously described [20, 21]. Glass slides were pre-cleaned and silanized with a 2% (v/v) 3-(trimethoxysilyl)propyl methacrylate (Sigma-Aldrich, St. Louis, MO, USA) solution prepared in ethanol for 30 minutes on a shaker; subsequently, silanized glass slides were immersed in ethanol to rinse and were dried on a hot plate for 5–15 minutes at 110°C. A pre-polymer solution of acrylamide/bisacrylamide (w/v) (Sigma-Aldrich) was mixed in dH₂O at percentage ratios of 4/0.4 and 8/0.55 to achieve Young's moduli of 1 kPa and 25 kPa, respectively; these ratios were selected to obtain a similar porosity [24]. Pre-polymer solutions were mixed with a 20% (w/v) Irgacure 2959 (BASF Corp.; Ludwigshafen, Germany) photoinitiator solution prepared in methanol at a ratio of 9:1 and added to a silanized glass slide. A 22×60 mm cover glass was used to sandwich the 9:1 pre-polymer:photoinitiator solution, and this system was placed in a UV oven for treatment with 365 nm UV A light (240×10^3 J) for 10 minutes for polymerization. Cover slides were removed after polymerization, excess reagents were purged in dH₂O for 72 hours, and dehydration was achieved on a hot plate for 15 minutes at 110°C prior to microspotting of ECM proteins. Dehydration was necessary to ensure that microspotted ECM proteins are able to embed within the gel network (the physical entrapment of ECM protein within the gel cannot be facilitated when gels are hydrated). The apparent elastic moduli were confirmed using nanoindentation measurements on an Optics11 Life Piuma Nanoindenter (Optics11 Life, Amsterdam, Netherlands). Specifically, a tip radius and stiffness of 59 μm and 0.28 N/m, respectively,

were used. The procedure was performed using an indentation rate of 10 μm per 2 seconds, followed by a 1 second residence time and subsequent tip evaluation at the same rate.

ECM proteins were prepared in a deposition buffer containing 38% (v/v) glycerol (Sigma-Aldrich), 10.55 mg/ml sodium acetate (Sigma-Aldrich), 3.72 mg/ml EDTA (Sigma-Aldrich), and 10mg/ml CHAPS (Sigma-Aldrich) in 1X phosphate buffered saline (PBS, Corning) using a 384-well V-bottom microplate (USA Scientific, Ocala, FL, USA). A set of 10 individual ECM proteins were selected based on prominence in human liver matrisome [25, 26] as well as previous work with liver progenitors [17], primary human hepatic stellate cells [21], and PHHs [20]. These ECM proteins included collagen I (C1; Millipore, Burlington, MA, USA), collagen III (C3; Millipore), collagen IV (C4; Abcam, Cambridge, MA, USA), collagen V (C5; Abcam), decorin (DC; R&D Systems, Minneapolis, MN, USA), fibronectin (FN, Millipore-Sigma, Burlington, MA, USA), hyaluronic acid (HA; Lifecore Biomedical, Chaska, MN, USA), laminin (LN; Millipore), lumican (LU; Acro Biosystems, Newark, DE), and tenascin C (TC; R&D Systems). Single- and two-way ECM combinations were utilized in this study to keep the scope manageable. A final protein concentration of 250 $\mu\text{g}/\text{mL}$ in the deposition solution was utilized for printing (i.e., 125 $\mu\text{g}/\text{mL}$ per ECM protein for a two-way combination was utilized). Each protein composition was transferred from the 384-well microplate to the dehydrated PA gel using an OmniGrid Micro (Digilab, Holliston, MA, USA) automated microspotter in a pre-arranged array format. Circular ECM protein microspots, henceforth referred to as “ECM islands”, were 450 μm in diameter (based on the size of the SMPC Stealth microarray pins, ArrayIt, Sunnyvale, CA) with 1 mm center-to-center spacing. Fluorescent labeling techniques have been previously applied to ECM microarrays to verify protein retention [27, 28]. Rhodamine-labeled dextran microspots were also spotted along one edge of the array for orientation alignment in post-processing analysis.

2.2 Cell culture

Pre-differentiated iHeps were commercially sourced from Fujifilm Cellular Dynamics International (Madison, WI, USA). Specifically, iCell 2.0 human iPSC-derived hepatocytes were from a Caucasian female and programmed from dermal fibroblasts whereas myCell human iPSC-derived hepatocytes (# 01279) were from a Caucasian male and originally programmed from a blood mononuclear cell; these two donors above were used for ECM microarray studies. A third myCell human iPSC-derived hepatocyte donor (# 01177) was the gift of Dr. Paul Watkins of the University of North Carolina at Chapel Hill and was used only for multi-well plate validation studies. Cells were thawed, counted, and resuspended to a density of $\sim 0.25 \times 10^6$ cells/mL in seeding media consisting of a 1X Roswell Park Memorial Institute (RPMI) 1640 Medium (Thermo Fisher Scientific, Waltham, MA, USA) base supplemented with 15 mM HEPES [4-(2-hydroxyethyl)-1-piperazineethane-sulfonic acid] buffer (Corning Life Sciences, Tewksbury, MA, USA), 1% insulin-transferrin-selenous acid-linoleic acid-bovine serum albumin (ITS⁺; Corning), 1% penicillin/streptomycin, 7 ng/ml glucagon (Sigma-Aldrich), 0.1 μM dexamethasone (Sigma-Aldrich), 5 μM Y-27632 (ROCK inhibitor; Sigma-Aldrich), 2.5 ng/ml recombinant human oncostatin-M (OSM; R&D Systems), 2% B27 (Thermo Fisher Scientific), and 10 μM of small molecule, functional proliferation hit 2 (FPH2; Sigma-Aldrich) that was identified in a previous publication as

being important for long-term iHep culture [29]. Prior to cell seeding, microarrays were hydrated in a solution of 1X PBS with 1% penicillin/streptomycin under UV sterilization for ~30 minutes. Approximately 4 ml of the cell suspension was added per array in a four-chamber rectangular culture dish, and cells were allowed to attach to microspot islands for 12–18 hours. Microarrays were washed 3x with 1X Dulbecco's Modified Eagle's Medium (DMEM; Corning) supplemented with 1% penicillin/streptomycin to remove non-adherent cells and replaced with serum-free maintenance media (seeding media formulation above but without the Y-27632 small molecule). Culture medium changes were performed every ~48 hours, and cellular fixation was performed for select microarrays on days 1, 7, or 14 using a 4% (v/v) paraformaldehyde solution (Alfa Aesar, Haverhill, MA, USA) prepared in 1X PBS for 10 minutes at room temperature (RT). Fixed microarrays were washed thoroughly using 1X PBS and stored at 4°C prior to immunostaining and high content imaging analysis.

2.3 Immunofluorescence staining and imaging

For immunostaining, arrays were pretreated for 1 hour at RT with a blocking and permeabilization buffer consisting of 5% (v/v) donkey serum (Southern Biotech, Birmingham, AL, USA) and 0.3% (v/v) Triton X-100 (Amresco, Solon, OH, USA) in a 1X Phosphate Buffered Saline (PBS; Corning) base. Following 3x washes with 1X PBS, arrays were incubated for 1 hour at RT with 1:200 goat anti-human albumin (Abcam, Cambridge, MA, USA), rabbit anti-human alpha-fetoprotein (AFP; GeneTex, Irvine, CA, USA), and mouse anti-human cytochrome P450 3A4 (CYP3A4; GeneTex), which were prepared in a dilution buffer consisting of 1% (w/v) bovine serum albumin (BSA; Fisher Scientific, Hampton, NH, USA) and 0.3% (v/v) Triton X-100 in a 1X PBS base. Subsequently, arrays were washed with 1X PBS three times for 5 minutes each, followed by a final rinse in ddH₂O. Secondary antibodies were diluted to 4 µg/ml in dilution buffer, including Alexa Fluor™ (AF) 488 donkey anti-goat IgG (Thermo Fisher Scientific), AF 567 donkey anti-rabbit IgG (Thermo Fisher Scientific), and AF 647 donkey anti-mouse IgG (Thermo Fisher Scientific), and then incubated on arrays for 1 hour at RT. Arrays were washed three times with 1X PBS, mounted in Fluoromount-G with DAPI (4',6-Diamidino-2-Phenylindole, Dihydrochloride; Southern Biotech), and sealed using a 24×60mm Gold Seal™ Cover Glass (Electron Microscopy Sciences, Hatfield, PA, USA). Automated imaging was performed on an IX83 microscope (Olympus America, Center Valley, PA, USA) with a high sensitivity 4.2MP ORCA-Flash4.0 LT+ sCMOS camera (Hamamatsu, Skokie, IL, USA).

2.4 Quantitative image analysis

High content imaging analysis was achieved through several software programs augmented with custom scripts as previously described [20]. Images acquired at 16-bits were first converted to TIFF files using Olympus CellSens Software (Olympus America). Using Fiji software [30], images were processed to 8-bits, rotated, and aligned based on the rhodamine-labeled dextran microspots to achieve uniform orientations for ECM composition mapping. Next, aligned microarray images were cropped into single island images using MATLAB. Each image was subsequently processed using CellProfiler software to obtain mean intensity values for each marker of interest at a single-cell resolution; this was reported in arbitrary intensity units (a.i.u.) [31].

2.5 Hydrogel-conjugated multi-well plate studies

Hydrogel bound 96-Softwell™ plates were acquired with Easy Coat™ composition (proprietary to manufacturer) from Matrigen (Brea, CA) at 1 kPa, 4 kPa, and 25 kPa stiffnesses. Young's moduli were measured using a spherical tip (1 mm diameter) indentation and generation of a force-displacement curve by the manufacturer. The ECM compositions were pre-formulated in a deep-well plate to a concentration of 25 µg/ml total concentration (e.g., 12.5 µg/mL per protein in two-way combinations). The solution containing ECM proteins was coated to Softwell plates at 50 µL per well for 2 hours at 37°C and then washed twice with dH₂O prior to seeding cells. Hepatocyte suspension was prepared using the same methods as for microarray experiments above and seeded at a density of 50,000 cells per well with overnight attachment (12–18 hours). Non-adherent cells were washed 2x with culture medium and medium was changed and collected every 2 days for ~3 weeks in culture. For these studies, we used two donors of iHeps from Fujifilm Cellular Dynamics International (iCell and myCell #01177) as well as a PHH donor (lot TLQ, age 52, Caucasian, female) from BioIVT (Baltimore, MD). PHHs were thawed based on previously published methods [20] and seeded and cultured in the same way as the iHeps to enable direct comparisons, except that B27, FPH2, and OSM were not added to the PHH culture medium as these factors are relevant for iHep culture only [22].

2.6 Biochemical assays

Albumin was measured using a sandwich-based enzyme linked immunosorbent assay (ELISA, Bethyl Laboratories, Montgomery, TX) with horseradish peroxidase detection and 3,3',5,5'-tetramethylbenzidine substrate (TMB, Rockland Immunochemicals, Boyertown, PA). Urea concentration was measured from collected supernatants using diacetyl monoxime with acid and heat (Stanbio Labs, Boerne, TX) [32]. Absorbance of samples was read on the Synergy H1 multimode plate reader (Biotech, Winooski, VT).

CYP3A4 and CYP2C9 enzyme activities were measured after 1 hour incubation of cultures with luciferin-IPA (Promega Life Sciences, Madison, WI) or 3 hours incubation with luciferin-H (Promega), respectively, followed by processing of collected supernatants per manufacturer's recommendations; luminescence was quantified with the Synergy H1 multimode reader. CYP1A2 enzymatic activity was measured by incubating cultures for 3 hours with 5 µM 7-ethoxyresorufin (Sigma-Aldrich) and then quantifying the fluorescence of metabolite resorufin in the supernatants (excitation/emission 550/585 nm) using the Synergy H1 multimode reader.

2.7 Data analysis

Data was exported in Excel files and post-processed using R statistical software for data summarization, visualization, and statistical analysis. First, all single-cell intensity values were summarized for each island, and then islands were summarized for an individual condition (e.g., cells on ECM island containing FN and TC on 25 kPa PA substrates after 14 days of culture). Values are reported based on summary values consisting of two iHep donors unless otherwise stated, and replicate microarray experiments were conducted for a total of 4 to 15 islands per condition. Box and whisker plots and bar graphs were created using the ggplot2 package, while linear regression modeling was performed using

the lm package (C1 was used as the reference group). Statistics were performed using the ggpubr package, and $\alpha < 0.5$ was used to determine significance. Alternatively, data were graphed using GraphPad Prism software (San Diego, CA, USA) to create heatmaps, bar plots, and scatterplots. Bar plots were analyzed using a one-way analysis of variance (ANOVA) with Dunnett's multiple comparison tests. Scatterplots were analyzed for simple linear regressions, and R^2 values were reported to represent the goodness of fit. For down selection studies, a two-way ANOVA was performed. Error bars represent the standard error of the mean (SEM) unless otherwise indicated.

3. RESULTS

3.1 Cellular microarray platform supports iHep attachment over 2 weeks of culture

A high-throughput cellular microarray platform was adapted to appraise the maturation of two commercially available iHep donors (iCell and myCell # 01279) over 2 weeks of culture. PA gels with a tunable stiffness (i.e., Young's moduli) of 1 kPa and 25 kPa were fabricated on glass slides and robotically spotted with pre-formulated mixtures of recombinant ECM proteins. ECM islands were created using ten liver-inspired proteins in single- and two-way combinations (55 combinations total), including collagen I (C1), collagen III (C3), collagen IV (C4), collagen V (C5), decorin (DC), fibronectin (FN), hyaluronic acid (HA), laminin (LN), lumican (LU), and tenascin C (TC). Microarrays were fixed and analyzed following 1, 7, and 14 days of culture to assess the synergistic role of stiffness and ECM interactions on iHep maturation. Thus, 660 unique microenvironmental conditions (with 4–15 replicate islands per condition) were evaluated in the present study. Importantly, the microenvironmental cues selected here have been previously validated for PHHs [20], which provides a benchmark data set for comparisons. Cellular phenotypes were analyzed using immunostaining of 3 markers that are widely utilized for assessing hepatocyte functionality *in vitro*: albumin is a mature marker of the liver's synthetic ability, AFP is an immature form of albumin present in fetal hepatocytes only, and CYP3A4 is a major detoxification enzyme that is responsible for ~50% of ingested drugs [33]; these markers, including the ratio of albumin to AFP, are routinely utilized to appraise iHep maturity in different culture platforms [22].

The iHeps were able to attach on cellular microarrays long-term *in vitro*, as exemplified by representative images (Figure 1A). Notably, there were differences in island integrity and immunofluorescence intensity on day 7 across different conditions indicating cell attachment and expression were regulated by the underlying ECM composition and substrate stiffness. The 25 kPa substrates supported increased initial iHep attachment (mean of 394 cells/island) than the 1 kPa substrates (mean of 292 cells/island) following 1 day of culture (Figure 1B). A significant increase in iHep retention on 25 kPa substrates was observed over 2 weeks, indicating that stiffer substrates support better attachment than soft substrates, though attachment varies greatly with ECM composition as discussed below. When separating out the individual iHep donors, the iCell donor supported higher attachment than the myCell donor but trends across stiffnesses over time were similar (i.e., 25 kPa retained more cells over time) (Supplemental Figure 1A–B).

Several ECM combinations, all of which contained C4, supported 1.5- to 2-fold better cell attachment/retention than C1-only controls after 14 days on the 1 kPa substrates (C1,C4; C4,C5; C4,DC; C4,FN; C4,HA; C4,LN; C4,LU; and C4,TC) (Figure 1C), though such outcomes were not improved over C1 controls on the 25 kPa substrates (Figure 1D). Long-term iHep attachment/retention relative to C1-only controls was inhibited across substrates of both stiffnesses by C5 alone and several of its combinations with other ECM proteins, including C1, C3, DC, HA, LU, and TC, but not FN, LN, or C4. Several of the non-collagenous proteins, including DC, HA, LU, and TC, on their own and in combinations with each other inhibited iHep attachment/retention across substrates of both stiffnesses following 14 days of culture, with TC alone being the exception on 1 kPa substrates (i.e., similar attachment as C1). On the other hand, FN and its combinations with the aforementioned non-collagenous proteins significantly inhibited iHep attachment/retention on the 25 kPa substrates but not the 1 kPa substrates. The inclusion of LN on its own or in a mixture blunted the negative effects of several other ECM proteins (C5,LN; DC,LN; HA,LN; LN; LN,LU; and LN,TC) on iHep attachment/retention after 14 days of culture, such that trends were similar to C1-only. Lastly, a moderate correlation was observed between both iHep donors after the initial seeding ($R^2=0.4104$ for 1 kPa and $R^2=0.5212$ for 25 kPa) for cell density; however, cell attachment was not correlated between donors following 2 weeks of culture (Supplemental Figure 1C).

3.2 Regulation of iHep albumin and AFP by ECM stiffness and protein composition

Due to their partially pre-differentiated phenotype, iHeps expressed minimal albumin protein on day 1 of culture irrespective of the underlying ECM (Figure 2B). Maturation then increased on both stiffnesses over 2 weeks of culture, with a significant increase in albumin expression on 25 kPa substrates at day 7 of culture as compared to the 1 kPa substrates (Figure 2B). The albumin expression observed for iHeps long-term on cellular microarrays was modulated by synergies with the underlying ECM composition. For instance, C1,HA supported a moderate albumin expression in iHeps on 25 kPa substrates and high albumin expression on 1 kPa substrates, whereas LN supported a moderately high albumin expression independent of stiffness (Figure 2A). Interestingly, 1 kPa substrates led to a significant upregulation in albumin expression in iHeps as compared to 25 kPa substrates after 14 days of culture (Figure 2B). These trends were consistent across both iHep donors (Supplemental Figure 2A–B). Heatmaps demonstrated the ECM-dependent regulation of albumin expression over 2 weeks of culture for all tested ECM combinations (Figure 2C). After 7 days of culture, C4 and its mixture with several other ECM proteins (C1, C3, C5, DC, FN, HA, LN, LU, and TC) caused higher albumin expression than C1-only controls across substrates of both stiffnesses. After 14 days of culture, the top ten ECM conditions with respect to high albumin expression relative to C1 controls on the 1 kPa substrates were LN; FN,TC; C5,DC; C3,HA; C5; FN; DC,TC; C5,TC; DC,HA; and C5,HA. On the 25 kPa substrates, several ECM conditions led to higher albumin expression than C1-only controls after 14 days of culture, with the top conditions being C3,HA; C5,DC; C5,FN; C5,HA; DC,HA; DC,LN; FN; FN,LN; FN,LU; FN,TC; HA,LN; LN; LN,LU; and LN,TC.

To further confirm the top-performing conditions, a linear regression analysis was performed to determine the contribution of each ECM composition to albumin expression relative to C1-only (Figure 2D). LN and FN,TC significantly upregulated iHep albumin expression on the 1 kPa substrates, whereas several additional ECM combinations (LN; FN,TC; C5,DC; C3,HA; C5; FN; DC,HA; C5,HA; FN,LN; FN,LU; LN,LU; C5,FN; C4,HA; and LN,TC) significantly upregulated albumin expression on the 25 kPa substrates. The iHeps did not show a correlation between albumin expression and cell density, except for a moderately positive correlation observed on day 7 of culture for the 1 kPa substrates only (Supplemental Figure 2C).

AFP was similarly assessed as an immature marker, given that it is found in fetal liver tissue. Herein, we observed positive AFP expression in iHeps following 1 day of culture (significantly increased on 1 kPa substrates) with an upregulation over time as with albumin (Figure 3A). As with albumin, the 1 kPa substrates promoted increased average expression of AFP by day 14 of culture and a downregulation on the 25 kPa substrates, likely due to a reduced maturation state of the cells (i.e., albumin and AFP are co-modulated on substrates). Representative islands for C1,DC and C4,DC showed an upregulation of both albumin and AFP by day 14 of culture, with a higher expression of albumin than AFP observed on the edge of the islands in the merged image (Figure 3B). The ratio of ALB to AFP is indicative of a maturing phenotype [4] and was thus evaluated here (Figure 3C). After 14 days of culture on the 1 kPa substrates, several ECM conditions led to 2 or higher ALB:AFP, including C5,HA; FN,TC; C5,TC; C3; FN,LU; LN; LU; C1,DC; TC; FN,LN; and C3,LN. After 14 days of culture on the 25 kPa substrates, ECM conditions that led to 2 or higher ALB:AFP included C5,HA; FN,TC; C5,TC; C5,DC; and LN,TC. However, donor-to-donor differences were observed with ALB:AFP ratios, such that the ratio was not significantly modulated by ECM conditions in the iCell donor (i.e., less than or equal to 1) though cells displayed higher ratios on the 25 kPa substrates as compared to the 1 kPa substrates. In contrast, the ratios were highly dependent on the underlying ECM composition for the myCell donor, reaching values of 3 and higher for several ECM conditions, including FN,TC; C5,HA; C5,TC; FN,LU; LN; C3; TC; C1,DC; C3,DC; HA,LN; and C3,LN (Supplemental Figure 3A). Overall, ~47% and ~33% of ECM combinations caused 1.3 or higher ALB:AFP ratios on 1 kPa and 25 kPa microarrays, respectively (Supplemental Figure 3B–C).

3.3 Regulation of iHep CYP3A4 expression by ECM stiffness and protein composition

Due to their partially pre-differentiated phenotype, iHeps expressed minimal CYP3A4 protein on day 1 of culture irrespective of the underlying ECM (Figure 4B). Maturation then increased up to 6-fold on both stiffnesses over 2 weeks of culture, with a significant increase in CYP3A4 expression on 25 kPa substrates at day 7 of culture as compared to the 1 kPa substrates. However, by 14 days in culture, average CYP3A4 expression was higher on the 1 kPa substrates versus the 25 kPa ones. Such a trend was similar across both iCell and myCell donors after 7 days of culture, but by day 14 of culture, the myCell donor displayed higher CYP3A4 expression on the 25 kPa substrates (Supplemental Figure 4A–B). The CYP3A4 expression in iHeps on cellular microarrays was further modulated by synergies with the underlying ECM composition. For instance, C1,HA supported long-term upregulation of

iHep CYP3A4 expression independent of substrate stiffness, whereas CYP3A4 intensity was higher on 1 kPa than 25 kPa for the C5,TC combination (Figure 4A). Heatmaps further revealed the effects of ECM stiffness and protein composition on CYP3A4 expression over 2 weeks of culture for all tested ECM combinations (Figure 4C). After 7 days of culture, C4 and its mixture with several other ECM proteins (C1, C3, C5, DC, FN, HA, LN, LU, and TC) caused higher CYP3A4 expression than C1-only controls across substrates of both stiffnesses. Additionally, C1,FN; C3,FN; FN; C5; FN;LN; LN; and LN,TC also caused higher CYP3A4 expression than C1-only controls across substrates of both stiffnesses. After 14 days of culture, the ECM conditions with respect to high CYP3A4 expression relative to C1-only controls on the 1 kPa substrates were C3,HA; C4; C5; C5,DC; C5,FN; DC,TC; and FN. On the 25 kPa substrates, several more ECM conditions led to higher CYP3A4 expression than C1-only controls after 14 days of culture, with the top conditions being C3,HA; C4,HA; C5,HA; DC,HA; FN,LN; HA,LN; LN; and LN,LU.

To further confirm the top-performing conditions, a linear regression analysis was performed to determine the contribution of each ECM composition to CYP3A4 expression relative to C1-only controls (Figure 4D). FN and FN,TC significantly upregulated iHep CYP3A4 expression on the 1k Pa substrates following 14 days of culture, while several ECM conditions caused significant downregulation, including HA; LU; DC,LN; DC,LU; LU,TC; HA,TC; HA,LN; C3,LU; LN,TC; C5,LU; and FN,HA. On the 25 kPa substrates after 14 days of culture, iHep CYP3A4 was significantly upregulated by C3,HA; C4,HA; DC,HA; LN; FN,LN; and LN,LU, whereas no significant CYP3A4 downregulation was observed due to ECM composition on the 25 kPa substrates. Lastly, CYP3A4 was moderately and positively correlated with cell density, but only for 1 kPa substrates after day 7 (both donors) and day 14 (myCell only) of culture (Supplemental Figure 4C).

3.4 Down-selection of ECM combinations that positively regulate iHep maturation

Next, cross analysis between long-term attachment/retention and hepatic markers was performed to down-select optimal microenvironmental combinations for iHep maturation. The first consideration was the retention of at least 100 iHeps/island over 2 weeks of culture towards providing statistical robustness in a single-cell analysis pipeline. Next, ECM conditions that promoted the highest expression of both albumin and CYP3A4 were down-selected, as shown by scatter plots for the 1 kPa (Figure 5A) and 25 kPa microarrays (Figure 5B). Due to the transient modulation of iHep phenotypes, only conditions that led to high and stable expression of mature markers from day 7 to 14 or that reached high-intensity levels by day 14 of culture were considered; specifically, these are labeled on the scatter plots to assess time-dependent changes from day 7 to day 14 (day 1 was omitted due to low functional levels) for albumin expression on 1 kPa (Supplemental Figure 5A) or 25 kPa (Supplemental Figure 5B) substrates and CYP3A4 expression on 1 kPa (Supplemental Figure 5C) or 25 kPa (Supplemental Figure 5D) substrates. Overall, the highest performing conditions for down-selection supported >0.15 arbitrary intensity units (a.i.u.) for both albumin and CYP3A4. For 1 kPa substrates, this included LN; C4; C3,LN; HA,LN; FN,LN; LN,TC; and TC, the majority of which relied on the inclusion of LN as a key driver of hepatic maturation on liver-like substrates. LN was similarly a positive promoter of hepatic

functions on 25 kPa substrates, for which top-performing ECM combinations included C3,HA; LN,LU; FN,LN; LN; LN,TC; and C5,DC.

3.5 Comparison of iHep and PHH responses on cellular microarrays

We have previously used cellular microarrays containing the same ECM combinations and stiffnesses with PHHs to identify top performing conditions for long-term *in vitro* maintenance [20]. Thus, here we compared responses in iHeps with PHHs on the microarrays. Overall, attachment on the microarrays was correlated between PHHs and iHeps on day 14 of culture for 1 kPa ($R = 0.6$, $p = 1e-6$) and 25 kPa ($R = 0.72$, $p = 4.9e-10$) substrates (Figure 6A–B). Several ECM conditions led to nearly identical attachment/retention across PHHs and iHeps on the 1kPa substrates (>100 cells/island on C4,LU; C4,LN; C1,C4; C3,FN; C3,C4; C4,C5; C4,HA; C4; and C1,FN) and on the 25 kPa substrates (>100 cells/island on C4,LU; C3,FN; C3,C4; C3,DC; and C4,C5). On the other hand, several ECM conditions led to high iHep attachment/retention (>100 cells/island) but not for PHHs (< 100 cells/island) over 14 days of culture on the 1 kPa substrates (HA,LN; LN; LN,TC; DC,LN; C5,LN; LN,LU; FN,LN; FN,HA; TC; C3,LN; and C1) and on the 25 kPa substrates (LU and HA,LN). Reciprocally, several ECM conditions led to high PHH attachment/retention but not for iHeps over 14 days of culture on the 1 kPa substrates (C3,DC; C1,DC; DC,HA; and C1,LU) and on the 25 kPa substrates (DC,FN; C5,LU; C5,TC; C5,HA; FN; and DC,HA). These results underscore the need to evaluate each type of hepatocyte independently on ECM conditions.

Albumin expression in iHeps and PHHs on different ECM compositions did not correlate on the 1 kPa substrates after 14 days of culture (Figure 6C), whereas a moderate positive correlation was observed on 25 kPa substrates (Figure 6D). Generally, albumin expression in the iHeps after 14 days of culture was 50–60% of PHH expression on average, but there were several ECM combinations that caused relatively high albumin expression in both cell types on the 1 kPa substrates (e.g., >0.2 a.i.u. on DC,TC; C3,HA; FN,TC; and C5) and the 25 kPa substrates (e.g., >0.15 a.i.u. on LN,TC; LN; DC,HA; C3,HA; FN,LN; and C5,HA). Interestingly, LN led to nearly 2-fold higher albumin expression in iHeps than in PHHs after 14 days of culture on the 1 kPa substrates, whereas there were several ECM combinations across both stiffnesses that caused higher albumin expression in PHHs than in iHeps, such as C3,C4; C1,TC; C4,FN; and C4,DC on 1 kPa substrates and in addition to these, C1,DC; C3,DC; C4,LN; C4,C5; C3,FN; and C1,HA on the 25 kPa substrates. The ratio of iHep to PHH albumin expression for all tested ECM conditions across the 1 kPa and 25 kPa microarrays is depicted visually in Supplemental Figure 6.

CYP3A4 expression in iHeps and PHHs on different ECM compositions displayed a moderate positive correlation on both 1 kPa (Figure 6E) and 25 kPa (Figure 6F) substrates. Interestingly, CYP3A4 expression reached up to 3-fold higher in iHeps after 14 days of culture than in PHHs across both stiffnesses. Nonetheless, several ECM combinations caused relatively high CYP3A4 expression in both cell types on the 1 kPa substrates (e.g., >0.15 a.i.u. for iHeps and >0.1 a.i.u. for PHHs on C5; C4,HA; C1,C4; C3,C4; C4,C5; and C1,FN) and the 25 kPa substrates (e.g., >0.15 a.i.u. for iHeps and >0.1 a.i.u. for PHHs on C4,HA). Lastly, several ECM combinations caused high CYP3A4 expression in iHeps

but very low expression in PHHs on the 1 kPa substrates (e.g., >0.15 a.i.u. for iHeps and <0.07 a.i.u. for PHHs on FN,LU; C3; DC; LN; and FN,LN) and on the 25 kPa substrates (e.g., C1,C5 and LN,TC). The ratio of iHep to PHH CYP3A4 expression for all tested ECM conditions across the 1 kPa and 25 kPa microarrays is presented visually in Supplemental Figure 7.

3.6 Evaluation of iHep functions within hydrogel-conjugated multiwell plates

The cellular microarray platform is a powerful tool for exploring in a high-throughput format the synergies of ECM composition and stiffness on cell phenotype using automated high-content imaging. However, microarrays do not allow interrogation of non-imaging based (i.e., biochemical) end-points due to shared media across the various ECM combinations, which necessitates supplementation of microarray data with multiwell plate-based culture formats. Towards that end, we first selected 8 ECM combinations from the microarray studies that led to retained cells (>100 cells/island for 1 kPa and >200 cells/island for 25 kPa substrates) and high expression of both albumin and CYP3A4 after 14 days of culture relative to previously published PHH phenotype on microarrays [20]. These 8 ECM combinations were then adsorbed to commercially available Matrigen™ plates at a constant total coating density for all combinations and at 1, 4, and 25 kPa stiffnesses. Two iHep donors (iCell and myCell #01177) and one PHH donor (TLQ) were cultured on the plates for ~3 weeks and several functions were assessed, including albumin and urea secretions in supernatants, and the activities of CYP3A4, 2C9, and 1A2 via incubation of the cells with prototypical substrates.

Albumin secretion (Figure 7A) was statistically similar across all 3 stiffnesses and the 8 down-selected ECM conditions for ~3 weeks of iHep (iCell) culture. Urea secretion (Figure 7B) was statistically similar across all 3 stiffnesses and the 8 down-selected ECM conditions for 15 days in culture, but at days 19 and 23 days of culture, urea secretion on 4 kPa was slightly lower than on the other two stiffnesses for specific ECM (e.g., C3,LN). In contrast, we observed significant stiffness and ECM composition dependent differences in CYP activities in the iHeps (Figure 7C). On the 1 kPa substrates, C1,LN; C3,LN; and C4 led to relatively high CYP2C9 and CYP3A4 activities while C5,FN; C4,LU and FN,LN led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Figure 7C, left). On the 4 kPa substrates, C1; LN; and C1,LN led to relatively high CYP2C9 and CYP3A4 activities while C4,LU and C5,FN led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Figure 7C, middle). Lastly, on the 25 kPa substrates, C1; LN; C1,LN; and C3,LN led to relatively high CYP2C9 and CYP3A4 activities while FN,LN and C5,FN led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Figure 7C, right). CYP trends in iHeps (iCell donor) on the 1, 4, and 25 kPa substrates after 7 and 14 days of culture are displayed in Supplemental Figure 8.

For the second iHep donor (myCell #01177), albumin and urea secretions were also statistically similar across the 3 tested stiffnesses and 8 down-selected ECM combinations (Supplemental Figure 9A,B). On the 1 kPa substrates, C3,LN; C1,LN and C4 led to relatively high CYP2C9 and CYP3A4 activities while C3,LN; C4,LU; and C5,FN led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Supplemental Figure 9C, left).

On the 4 kPa substrates, C3,LN; C1; C1,LN; and C4 led to relatively high CYP2C9 and CYP3A4 activities while C3,LN and C4,LU led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Supplemental Figure 9C, middle). Lastly, on the 25 kPa substrates, LN; C4; and C1,LN led to relatively high CYP2C9 and CYP3A4 activities while C1; C5,FN; and FN,LN led to relatively high CYP1A2 activity in the iHeps after 3 weeks (Supplemental Figure 9C, right). CYP trends in iHeps (myCell #01177) on the 1, 4, and 25 kPa substrates after 7 and 14 days of culture are displayed in Supplemental Figure 10.

3.7 Comparison of iHep and PHH morphology and functions within hydrogel-conjugated multiwell plates

When comparing iHep functions in the hydrogel-conjugated multiwell plates to those in PHHs across down-selected ECM combinations (C1,LN; C3,LN; C4,LU; and C5,FN), we observed that iHep cultures from both donors secreted ~3- to 12-fold more albumin than PHH cultures after 1 week, and while albumin secretion rates were relatively stable in the iHep cultures between 7 and 19 days, rates declined continuously in the PHH cultures (Figure 8A and Supplemental Figure 11A). In contrast, PHH cultures secreted ~2- to 9-fold more urea than iHep cultures from both donors after 1 week; however, urea secretion rates continuously declined in PHH cultures while increasing over time in iHep cultures, and by ~3 weeks, iHep cultures secreted ~2- to 20-fold more urea than the PHH cultures (Figure 8B and Supplemental Figure 11B). For CYP1A2 activity, iHep cultures from both donors displayed ~7- to 55-fold higher activities after ~3 weeks than PHH cultures for the above ECM combinations (Figure 8C and Supplemental Figure 11C). For CYP2C9 activity, iHep cultures from both donors displayed ~5- to 448-fold higher activities after ~3 weeks than PHH cultures for the above ECM combinations (Figure 8D and Supplemental Figure 11D). For CYP3A4 activity, iCell iHep cultures displayed ~3- to 9-fold higher activities after ~3 weeks than PHH cultures for the above ECM combinations (Figure 8E), but for the myCell (#01177) iHep cultures, CYP3A4 activities were similar to the PHH cultures after ~3 weeks (Supplemental Figure 11E).

The functional kinetics across iHep and PHH cultures discussed above could be due to a combination of proliferation (iHeps only as PHHs are non-proliferative under these conditions at least), apoptosis, and/or functional changes on a per cell basis. Phase contrast images revealed that while iHeps remained attached to even 1 kPa substrates for the most part over 3 weeks, PHH numbers declined over time irrespective of substrate stiffness and ECM protein coating (Supplemental Figure 12). Furthermore, while the iCell iHeps tended to remain as monolayers on the down-selected ECM combinations, the myCell (#01177) iHeps formed hemi-spherical 3D structures over time, especially on the 1 kPa substrates coated with C4,LU and C5,FN.

4. DISCUSSION

We utilized ECM microarrays to elucidate for the first time how major liver ECM proteins regulate iHep maturation on varying stiffnesses over prolonged culture. Ten ECM proteins in single and two-way combinations were spotted robotically onto PA-coated glass slides into circular islands of 450 μm diameter, found to be sufficient for maintaining necessary

homotypic contacts between iHeps [22]. We chose 1 kPa (Young's modulus) to represent native liver stiffness and 25 kPa to mitigate any issues with cell attachment on very soft surfaces but still orders of magnitude lower than plastic or glass. High content imaging was used to assess iHep numbers (DAPI) and phenotype (albumin, AFP, and CYP3A4). These functional markers are used to assess the differentiation of iPSC-derived hepatoblasts into hepatocyte-like cells across different protocols [4], while their maintenance is used to assess hepatic phenotypic stability [1]. Staining intensities for albumin and CYP3A4 proteins correlate well with orthogonal assays (e.g., ELISA, qPCR) [12]. Additionally, AFP is the fetal form of albumin and is upregulated at the definitive endoderm [34] and hepatoblast [35] stages of lineage specification; albumin to AFP ratio, with values >1 indicating a maturing phenotype, is routinely utilized for appraising iHep maturity [36].

The iHeps remain better attached on 25 kPa than 1 kPa after 2 weeks which is likely due to a greater number of focal adhesions that allow the cells to attach better to stiffer surfaces due to stronger traction forces [37] while highly compliant surfaces reduce cell spreading and overall attachment [37, 38]. However, too much hepatocyte spreading on stiffer surfaces is correlated to de-differentiation [22, 39, 40], which thus necessitates an optimal balance between cell attachment and spreading for the induction of differentiated functions. iHep attachment was strongly influenced by ECM composition on both stiffnesses. C4 mixed with C1, C5, DC, FN, HA, LN, TC, and LU supported up to 2-fold higher iHep attachment relative to C1 after 2 weeks on 1 kPa but not 25 kPa. C5, DC, HA, LU, and TC inhibited iHep attachment on their own and in combination with each other across both stiffnesses, while LN on its own or in a mixture blunted the negative effects of the above proteins.

ECM protein composition and stiffness were also strong regulators of iHep functions on microarrays. While iHeps on 25 kPa upregulated functions at a faster rate than 1 kPa and cells achieved a peak for median albumin expression, AFP, and CYP3A4 at day 7 on 25 kPa, 1 kPa supported higher median expression of phenotypic markers over 2 weeks. Our iHep findings are consistent with previous literature that showed that immortalized liver stem cells [41], mouse ESC-derived hepatocyte-like cells [13], and human ESC-derived hepatocyte-like cells [14] displayed higher hepatic functions on softer, liver-like substrates than stiffer ones. However, iPSC-derived embryoid bodies differentiated into hepatocyte-like cells did not previously display higher functions on C1- or FN-coated PA gels (0.6 to 50 kPa) versus plastic [42]. In contrast, here we identified several ECM combinations that upregulated mature iHep functions over 2 weeks on 1 kPa (LN; C4; C3,LN; HA,LN; FN,LN; LN,TC; TC) and 25 kPa (C3,HA; LN,LU; FN,LN; LN; LN,TC; C5,DC). AFP was upregulated along with albumin in iHeps on microarrays, which is consistent with previous literature that showed that both markers are co-regulated as iHeps further mature *in vitro* [4, 22]. Therefore, evaluation of albumin to AFP ratio is more indicative of a higher iHep maturation state. Of the ECM combinations that retained at least 100 iHeps per island and led to high and stable expression of mature markers over 2 weeks, several combinations led to high ALB:AFP (>1.5) on 1 kPa (LN; C3,LN; FN,LN; TC) and 25 kPa (LN,TC; C5,DC) substrates.

Of the 7 optimal ECM combinations identified above, 5 include LN. LN supported cell attachment and significantly upregulated albumin expression over 2 weeks independent

of substrate stiffness. The LN we used here was a mixture of different isoforms and of the various isoforms, LN 411 was previously identified as a positive regulator of iHep differentiation relative to C1 [12], while LN 521 and 111 isoforms were also shown to increase hepatic functions of differentiated human ES cells than Matrigel [11]. Lastly, while LN and TC on their own supported high levels of iHep attachment, function, and ALB:AFP ratio on 1 kPa, their mixture was necessary to support such outcomes on 25 kPa, whereas the mixture caused little to no improvement in ALB:AFP ratio on 1 kPa.

Comparing iHep functions with PHHs is important to determine concordance and ascertain the potential utility of the iHeps as a more sustainable and personalized cell source for applications. Thus, we compared iHep responses obtained here with our previously published study using PHHs cultured on ECM microarrays with the same combinations and stiffnesses [20]. PHH and iHep attachment on microarrays correlated well over 2 weeks across both stiffnesses, CYP3A4 correlated moderately, while albumin expression was not highly correlated. C4 alone and in mixture with LU, LN, C1, C3 C5, and HA caused similar attachment across the cell types. LN alone and in combination with HA, TC, DC, C5, LU, FN, and C3 caused higher iHep attachment on 1 kPa over 2 weeks than for PHHs, whereas PHH attachment was higher on mixtures of DC with C1, C3, HA, and FN and on mixtures of C5 with LU, TC, and HA across both stiffnesses. For albumin, LN on its own or in a mixture with TC or FN led to relatively high expression in both PHHs and iHeps on 1 kPa with LN causing ~2-fold higher albumin expression in iHeps than PHHs over 2 weeks. In contrast, C4 mixed with C3, FN, and DC on 1 kPa and mixed with C5 and LN on 25 kPa caused higher albumin expression in PHHs relative to iHeps. ECM combinations containing C4 (C4; C4,LU; C4,HA; C4,C5 on 1 kPa and C4,HA on 25 kPa) or LN (LN; C1,LN; C3,LN; C5,LN; FN,LN; HA,LN on 1 kPa and LN,TC on 25 kPa) or FN (FN; C3,FN; C5,FN; DC,FN; FN,LU; FN,TC on 1 kPa and FN,LU on 25 kPa) or C5 (C5; C1,C5; C3,C5; C5,DC; C5,HA; C5,TC on 1 kPa and C1,C5 on 25 kPa) led to a ~2-fold increase in CYP3A4 expression in iHeps relative to PHHs over 2 weeks. The above functional differences across iHeps and PHHs may be due to donor-dependent differences, which is one disadvantage of using commercial cells from different vendors, though such allows others to replicate and use our findings with the same cells.

To complement and validate microarray studies, we appraised the functional output of iHeps and PHHs on 8 down-selected ECM compositions (good cell attachment and high albumin and CYP3A4 expression relative to PHHs from microarrays) adsorbed onto hydrogel-conjugated multiwell plates of 1 kPa, 4 kPa, and 25 kPa stiffnesses. Albumin and urea secretions were similar across all 3 stiffnesses for the ECM conditions for 2–3 weeks; however, CYP activities were highly dependent on these parameters. Across all three stiffnesses, C1,LN led to relatively high CYP2C9 and CYP3A4 activities (consistent with microarray data) while C5,FN led to relatively high CYP1A2 activity in iHeps after 3 weeks. Furthermore, iHep cultures secreted up to 12-fold and 20-fold more albumin and urea, respectively, and displayed up to 55-fold higher CYP1A2 activities, up to 448-fold higher CYP2C9 activities, and up to 9-fold higher CYP3A4 activities than the PHH cultures after 3 weeks; such differences across the cell types are likely due to the declining PHH numbers over time irrespective of substrate stiffness and ECM protein composition whereas iHeps remain attached for 3 weeks even on the 1 kPa substrates. However, further molecular

level characterization would need to be carried out in the future to fully decouple the effects of cell proliferation, apoptosis, and per cell functional changes on the overall behavior of iHep and PHH cultures on the chosen substrates. Nonetheless, our results here indicate that relatively straight-forward combinations of ECM stiffness and protein compositions can be used to induce high levels of differentiated functions in iHeps for several weeks, which should bode well for utility in downstream applications, such as drug screening.

While this study is the first to elucidate the synergistic effects of ECM protein composition and stiffness on human iHep attachment and differentiated functions, the liver microenvironment contains non-parenchymal cells and gradients of oxygen and other soluble factors (e.g., cytokines, hormones, nutrients) that likely synergize with ECM to regulate hepatic functions across the functional liver zones [43]. ECM microarrays are modular to evaluate synergies between ECM and other microenvironmental cues via the seeding of multiple cell types on the ECM islands and inclusion within flat-plate bioreactors [44] to induce gradients of soluble factors onto the various cell-laden ECM islands. Additionally, cells on microarrays can be incubated with small molecular modulators of mechanosensitive pathways and subjected to traction force microscopy to elucidate the mechanisms underlying the observed phenotypic responses [17, 21]. Lastly, translating our findings to 3D gel based constructs will be useful to model the viscoelastic properties of liver tissue in both physiology and cell-dependent changes in fibrosis [45].

In conclusion, ECM microarrays were utilized to elucidate the synergistic role of ECM protein composition and stiffness on human iHep phenotype over several weeks in culture. In the absence of detailed mechanistic insights into the various molecular regulators of iHep differentiation, such high-throughput investigations of the role of distinct and combinatorial microenvironmental cues are critical for optimizing human liver platforms for drug screening and regenerative medicine, and to elucidate the role of ECM components and their interactions in liver physiology and disease.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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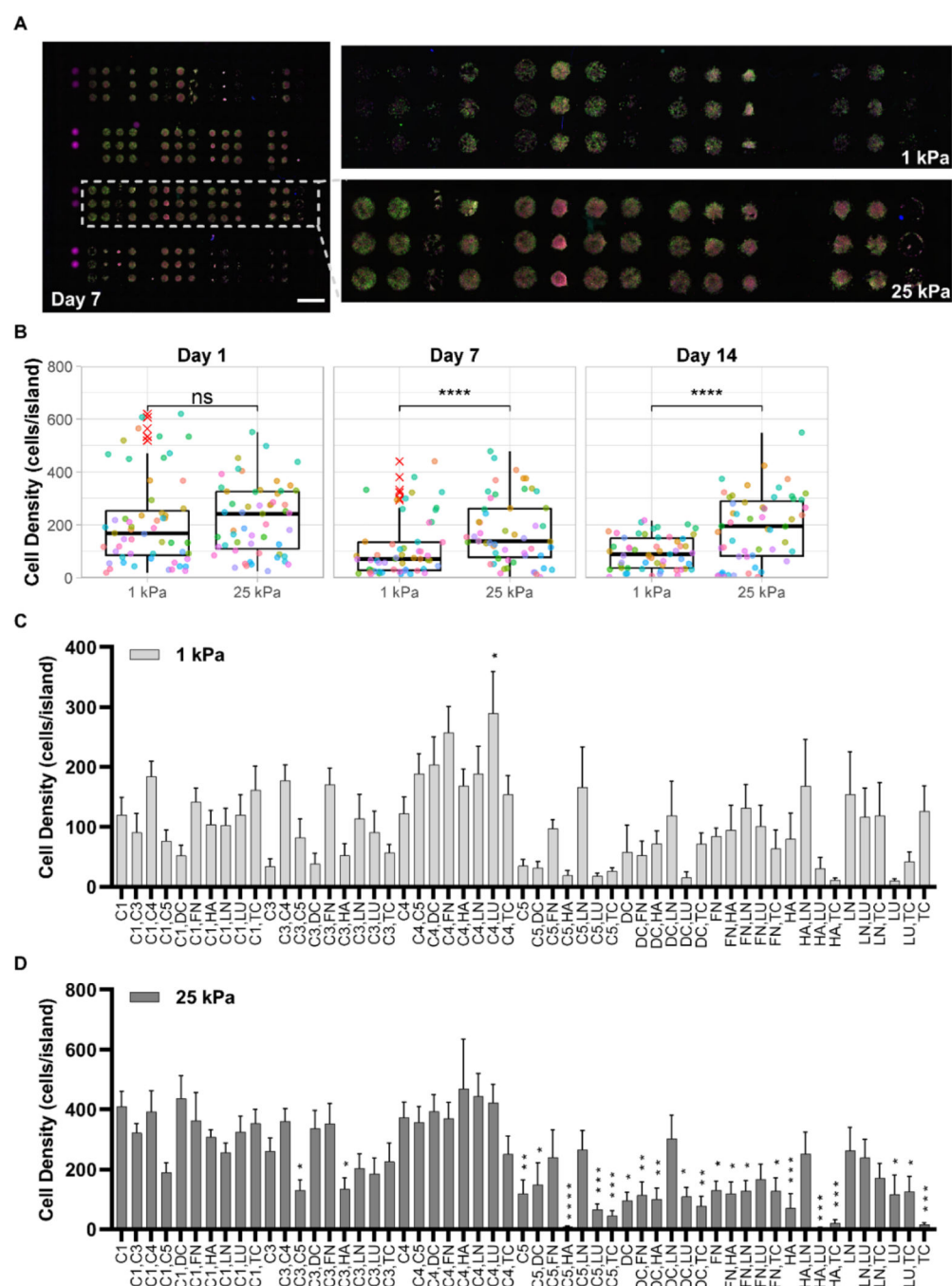


Figure 1: iHep attachment on cellular microarrays.

A) A 4-channel composite representative image of a complete cellular microarray for 25 kPa substrates on day 7 of culture is shown, and a subset of microspot 'islands' are shown on the left for 1 kPa (top left) and 25 kPa (bottom left); DAPI = blue, albumin = green, alpha-fetoprotein = orange; CYP3A4 = red. Scale bar = 2 mm. **B)** Box and whisker plots demonstrate the median cell density (center bar) and the interquartile range over time; each open circle represents a single ECM combination, and a red asterisk represents outliers beyond 1.5*IQR. Wilcoxon test was performed for mean comparisons

with 'ns' denoting $p > 0.05$ and $****p < 0.0001$. **C)** Bar chart represents cell density for each ECM combination after 14 days of culture for 1 kPa (top) and 25 kPa (bottom) substrates. Error bars represent the standard error of the mean. A one-way analysis of variance was performed with Dunnett's multiple comparisons against the C1 control with $*p < 0.05$, $**p < 0.01$, $***p < 0.001$, and $****p < 0.0001$. For panels B and C, data is comprised of single-cell measurements summarized across two iHep donors per condition ($n = 4$ to 15 ECM islands per condition).

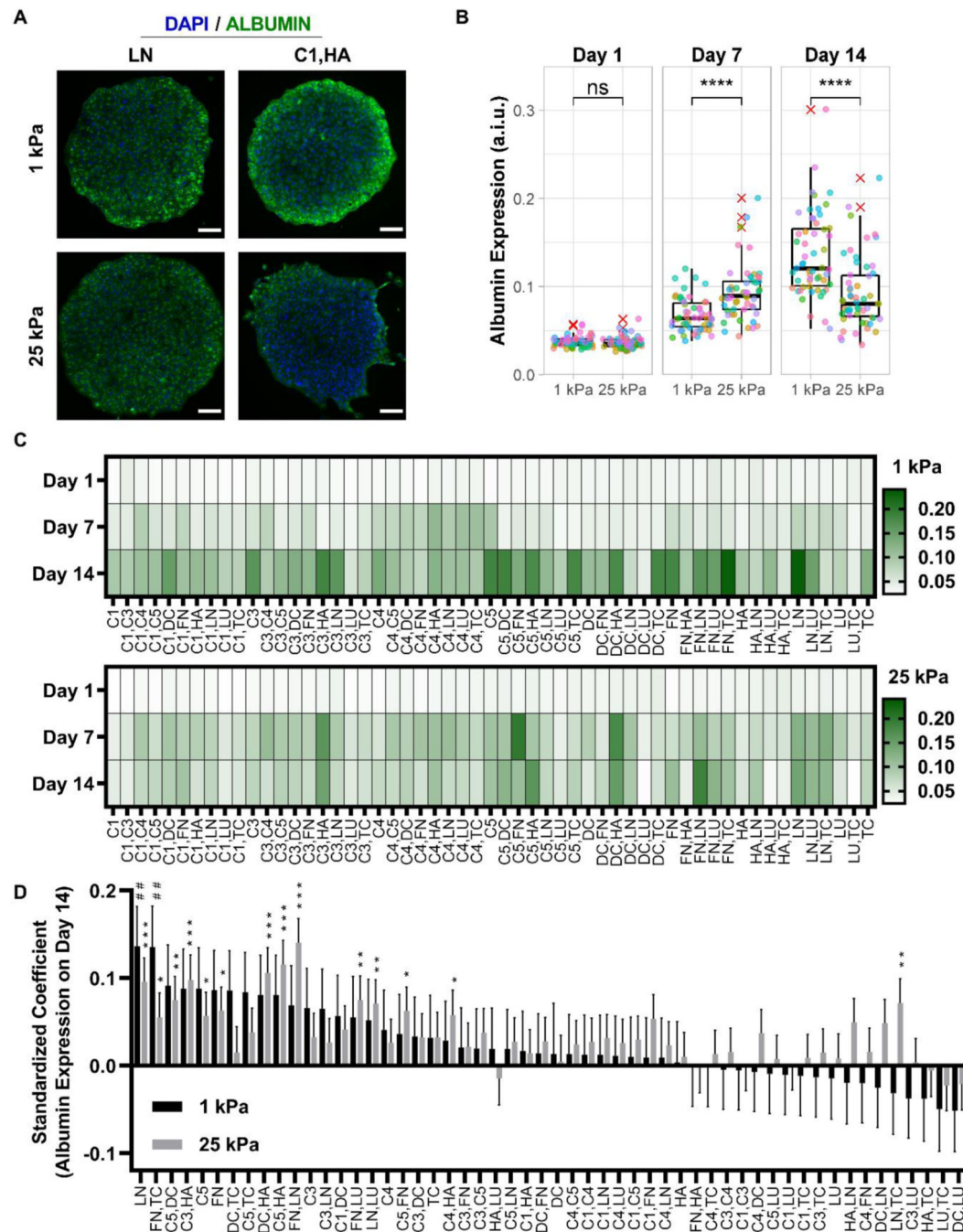


Figure 2: Albumin expression in iHeps on cellular microarrays.

A) Representative images for albumin expression. Scale bar = 100 μ m. **B)** Box and whisker plot plots demonstrate the median (center bar) and interquartile range for iHep albumin expression over time per each stiffness (1 kPa and 25 kPa); each open circle represents a single ECM combination, and a red asterisk represents outliers beyond 1.5*IQR. A.i.u. = arbitrary intensity units. Wilcoxon test was performed for mean comparisons with 'ns' denoting $p > 0.05$ and **** $p < 0.0001$. **C)** Heatmap of mean albumin expression for each individual extracellular matrix (ECM) composition over time for 1 kPa (top) and 25 kPa

(bottom) substrates. **D)** Rank ordering the effects of ECM composition and stiffness on albumin expression of iHeps using linear regression modeling with C1 as the reference condition (the x-axis represents the C1 reference, and the standardized coefficient for each ECM composition shows a positive or negative effect on the marker expression relative to C1). For 1 kPa, ^{##}p<0.001; for 25 kPa, *p<0.01, **p<0.001, and ***p<0.0001. For panels A, C, and D, data is comprised of single-cell measurements summarized across two iHep donors per condition (n = 4 to 15 ECM islands per condition).

A.i.u. = arbitrary intensity units. Wilcoxon test was performed for mean comparisons with ****p<0.0001. **B)** Representative images for albumin (ALB), AFP, DAPI (nuclear counterstain), and merge of all three channels for C1,DC, and C4,DC on 1 kPa and 25 kPa

substrates. Scale bars = 100 μm . **C)** Bar plot of the ALB:AFP ratio for 1 kPa and 25 kPa substrates on day 14 of culture. Error bars = standard error of the mean. For panels A and C, data is comprised of single-cell measurements summarized across two iHep donors per condition (n = 4 to 15 ECM islands per condition).

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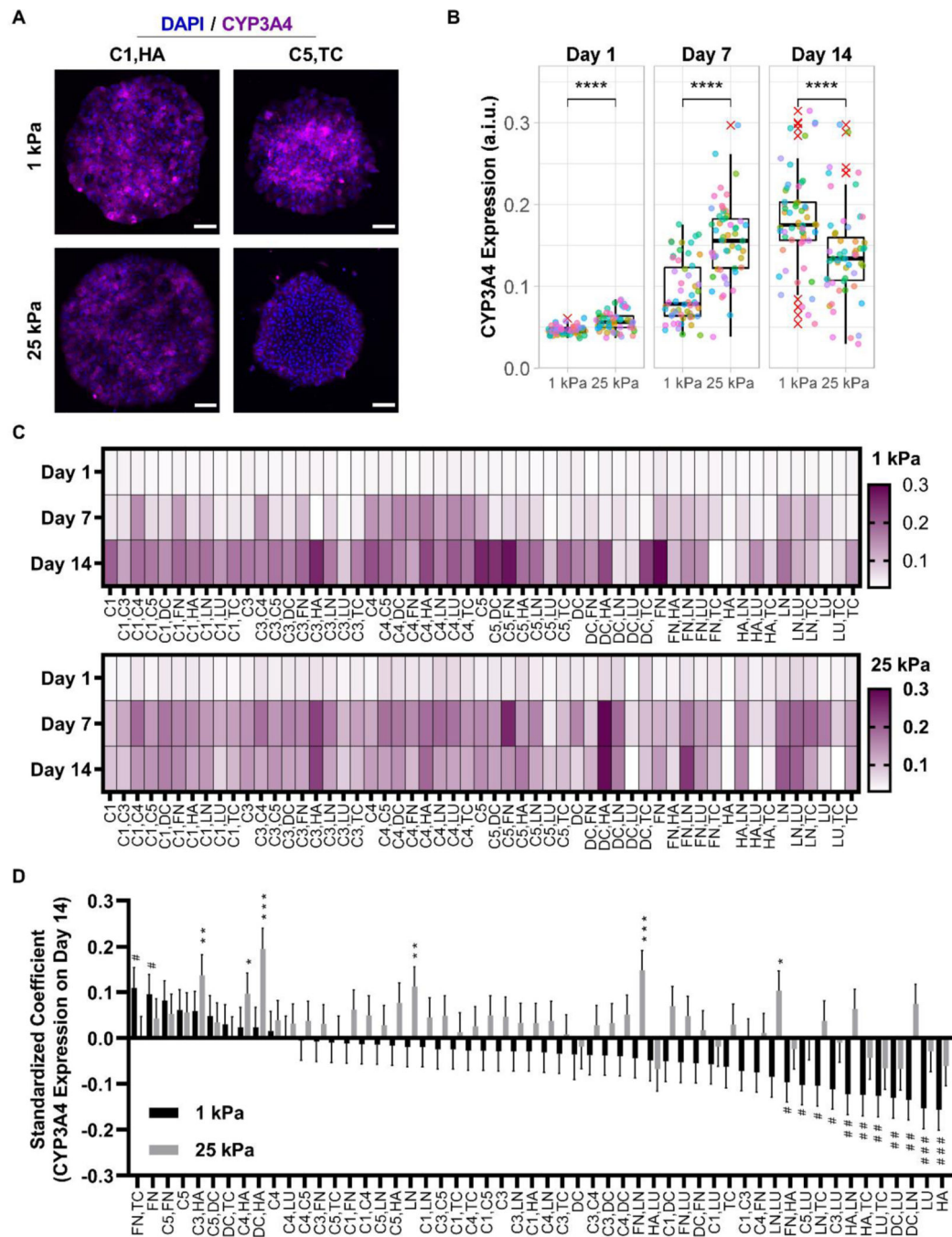


Figure 4: CYP3A4 expression in iHeps on cellular microarrays.

A) Representative images for CYP3A4 expression on day 14 of culture, including C1,HA, and C5,TC on 1 kPa and 25 kPa substrates. Scale bars = 100µm. **B)** Box and whisker plot plots demonstrate the median (center bar) and interquartile range for CYP3A4 expression over time per each stiffness (1 kPa and 25 kPa); each open circle represents a single ECM combination, and a red asterisk represents outliers beyond 1.5*IQR. A.i.u. = arbitrary intensity units. Wilcoxon test was performed for mean comparisons with ****p<0.0001. **C)** Heatmap of mean CYP3A4 expression for each individual extracellular matrix (ECM)

composition over time for 1 kPa (top) and 25 kPa (bottom) substrates. **D)** Rank ordering the effects of ECM composition and stiffness on CYP3A4 expression of iHeps using linear regression modeling with C1 as the reference condition (the x-axis represents the C1 reference, and the standardized coefficient for each ECM composition shows a positive or negative effect on the marker expression relative to C1). For 1 kPa, [#]p<0.01; for 25 kPa, *p<0.01, **p<0.001, and ***p<0.0001. For panels A, C, and D, data is comprised of single-cell measurements summarized across two iHep donors per condition (n = 4 to 15 ECM islands per condition).

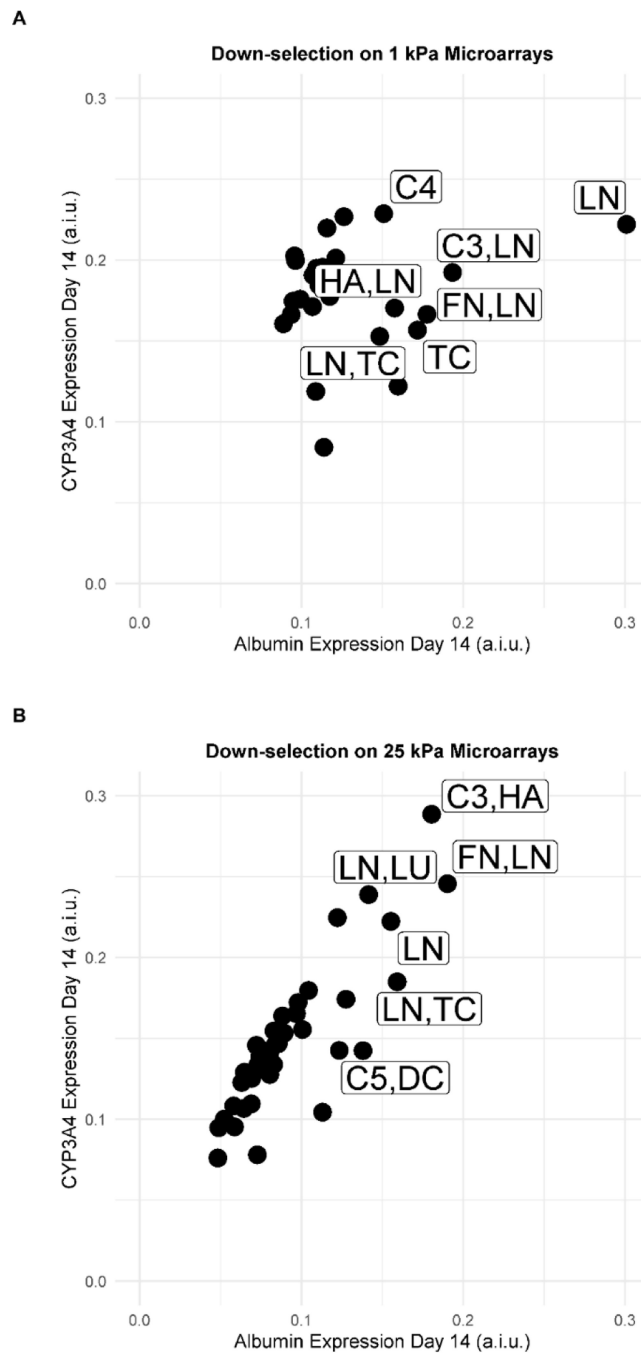


Figure 5: Down-selected conditions for iHep maturation on cellular microarrays.

A) Top conditions were identified that supported long-term expression of CYP3A4 (y-axis) and albumin (x-axis) on 1 kPa substrates and on **B)** 25 kPa substrates. A.i.u. = arbitrary intensity units. For all panels, data is comprised of single-cell measurements summarized across two iHep donors per condition (n = 4 to 15 ECM islands per condition).

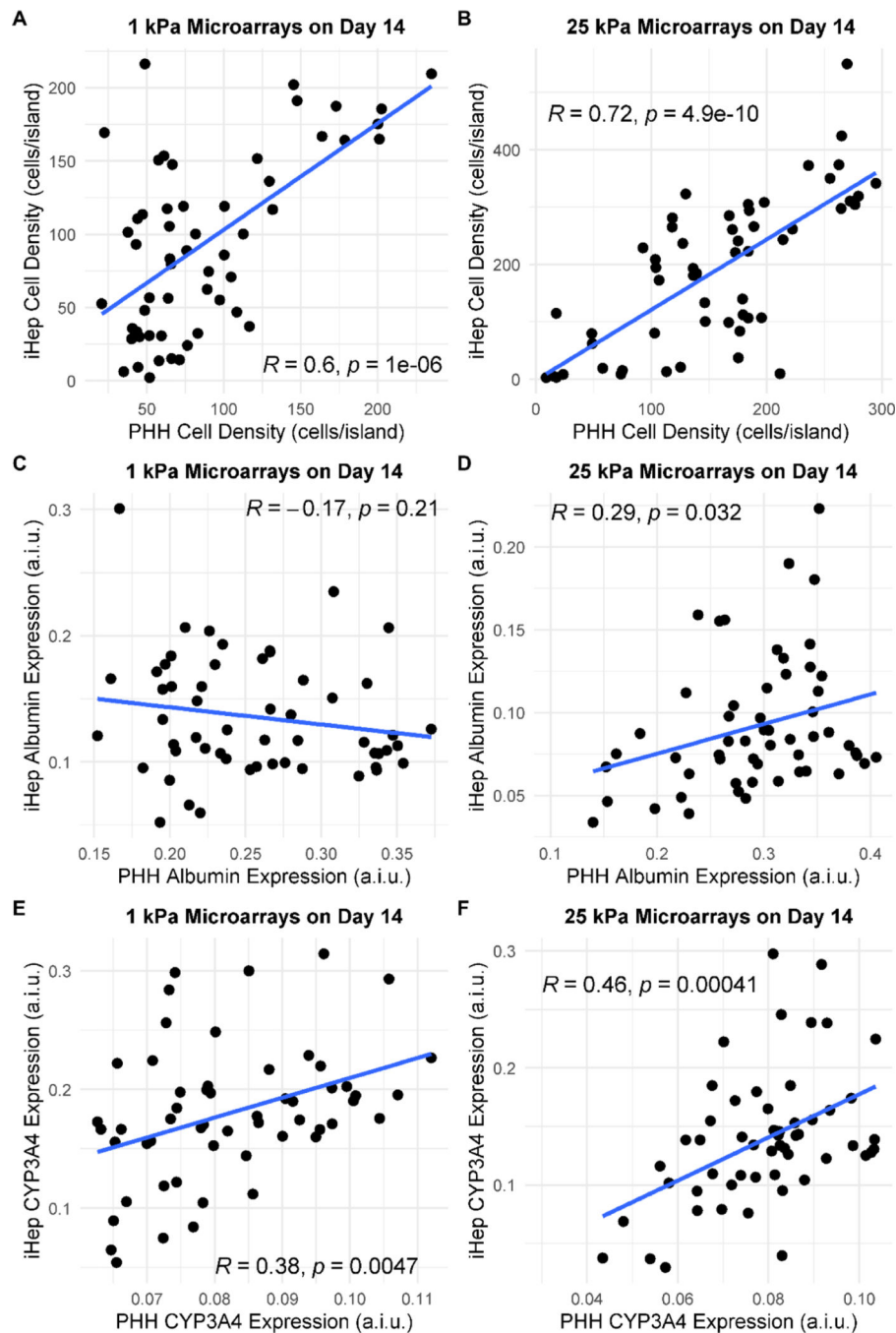


Figure 6: Comparisons of PHH and iHep functions on cellular microarrays.

A) Scatter plot showing the density of iHeps (y-axis) to PHHs (x-axis) for cell attachment per island on day 14 of culture on 1 kPa and **B)** 25 kPa substrates. **C)** Similarly, iHeps and PHHs are compared for albumin expression on 1 kPa and **D)** 25 kPa substrates, and for **E)** CYP3A4 expression on 1 kPa and **F)** 25 kPa substrates. A.i.u. = arbitrary intensity units. For all panels, iHep data is comprised of single-cell measurements summarized across two iHep donors per condition ($n = 4$ to 15 ECM islands per condition), whereas PHH data is

comprised of single-cell measurements summarized across three PHH donors per condition (n = 22–24 ECM islands per condition).

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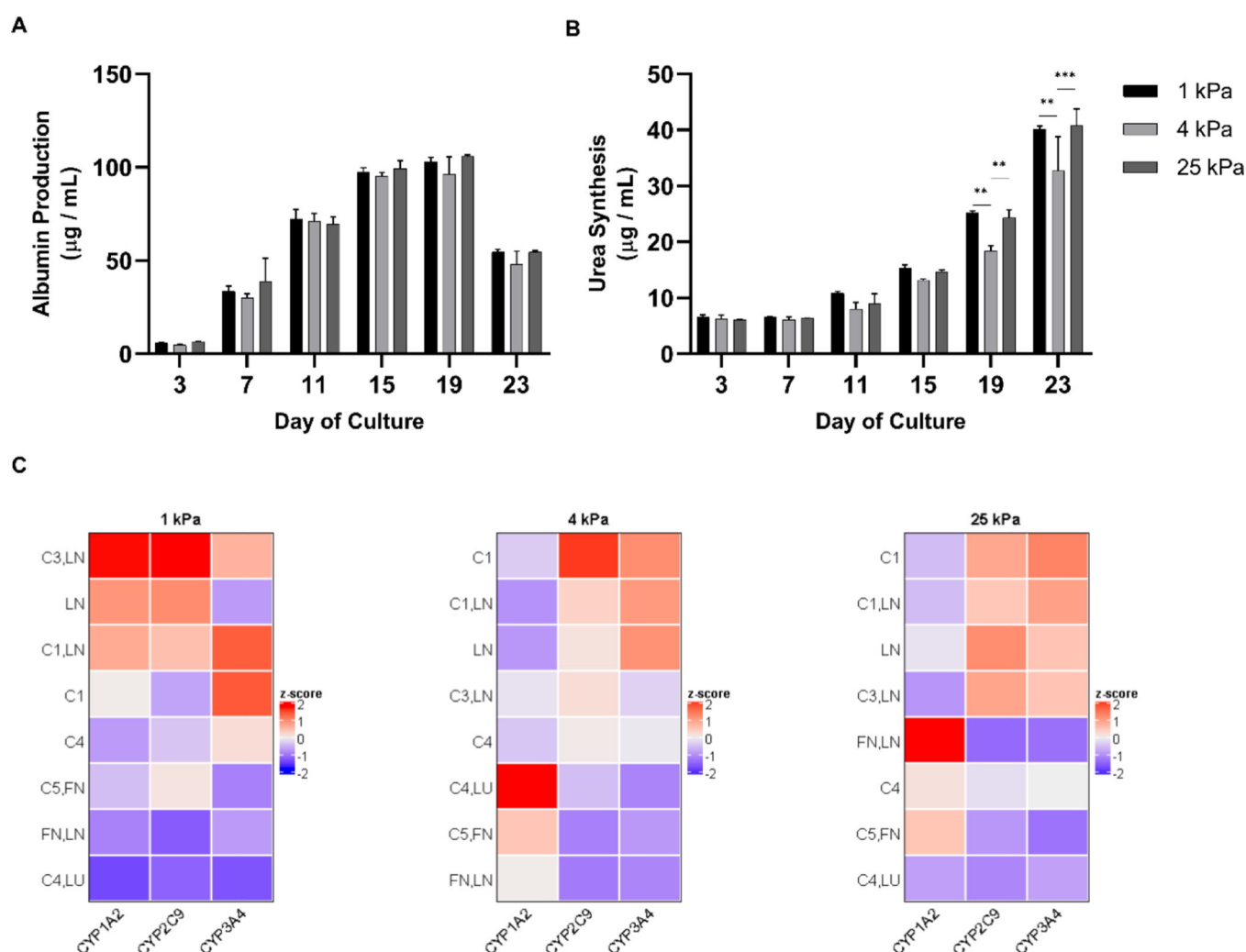


Figure 7: Functional assessment of iHeps on down-selected ECM combinations adsorbed to hydrogel-conjugated multiwell plates.

A) Albumin production and **B)** urea synthesis over time from iCell iHeps cultured on substrates of different stiffnesses with adsorbed C3,LN. For statistical analysis, a two-way ANOVA was performed with multiple comparisons at each timepoint; ** $p < 0.0021$, *** $p < 0.0002$. **C)** Heat maps showing activities of CYP1A2, CYP2C9, and CYP3A4 in iHeps after 3 weeks on 1 kPa (left), 4 kPa (middle), and 25 kPa (right) substrates with different adsorbed ECM compositions.

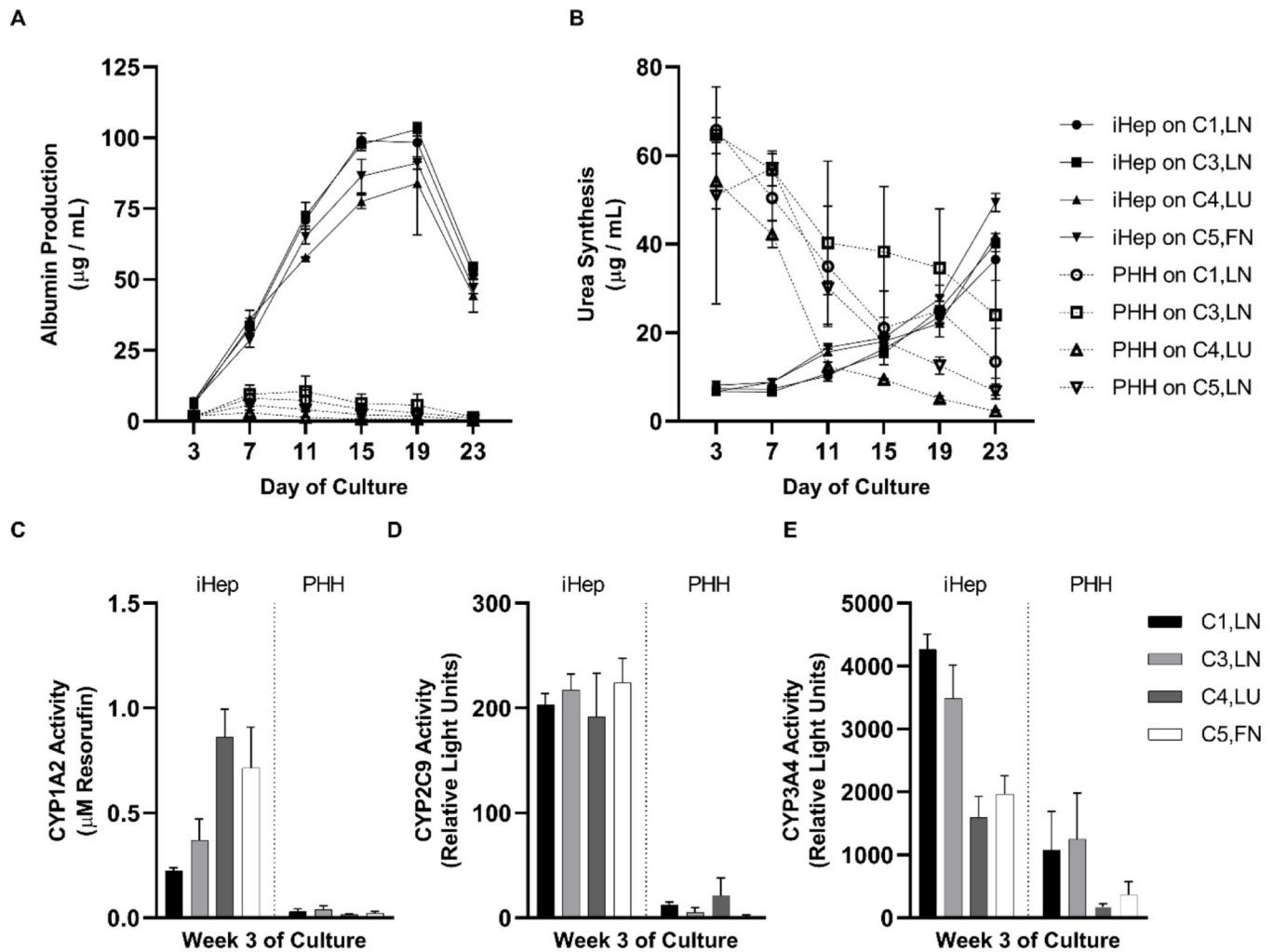


Figure 8: Long-term functions of iHeps versus PHHs on down-selected ECM combinations adsorbed to hydrogel-conjugated multiwell plates.

A) Albumin production and **B)** urea synthesis for iCell iHeps (closed shapes) and PHHs (open shapes) over time on 1 kPa substrates with different adsorbed ECM compositions. **C)** CYP1A2, **D)** CYP2C9, and **E)** CYP3A4 enzyme activities in iHeps (iCell donor) and PHHs after 3 weeks of culture on 1 kPa substrates with different adsorbed ECM compositions.